Back

Time to Practice: Density Estimation Methods (Part 1)
Graded Assignment • 15 min

Due Oct 16, 11:59 PM IST

Your grade: 100%

Your latest: **100%** • Your highest: **100%** • To pass you need at least 80%. We keep your highest score.

Next item →

1. What is the primary purpose of density estimation? 1 / 1 point

- (•) To compute the probability of observing an event in a specific region of the data space
- ( ) To identify the most frequent data point in a dataset
- ( ) To determine the average size of objects in a region

✓ Nice work

Density estimation is used to compute the probability of observing an event at a particular location based on a set of observations.

2. What is a key characteristic of a probability density function? 1 / 1 point

- ( ) It must be non-negative for all values
- (•) The total probability mass must add up to one over the domain
- ( ) It must have a maximum value of one
- ( ) It must sum to zero over the domain

✓ Nice work

A probability density function must integrate (add up) to one over the entire domain where observations are made, reflecting the total probability mass.

3. If a researcher needs to compute the probability of an event occurring within a specified area in a two-dimension space (e.g. a map), how would they make use of a density estimation over that space? 1 / 1 point

- ( ) By finding the maximum value of the density estimate in the region of interest
- (•) Integration of the probability density function over the region of interest
- ( ) Calculation of the cumulative distribution function at specific points

✓ Nice work

The probability of an event occurring within a specific area is calculated by integrating the probability density function over that area, as per the principles of density estimation.

4. What is the key difference between parametric and non-parametric density estimation? 1 / 1 point

- (•) Parametric density estimation assumes the data follow a specific parametrized probability distribution family (e.g. Gaussian), while non-parametric methods don't assume any particular underlying distribution.
- ( ) As the name suggests, parametric density estimators have specific parameters that can be tuned to adjust the fit of the density estimator, while non-parametric methods are parameter-free.
- ( ) Parametric density estimators estimate only the perimeter of probability density regions, while non-parametric density estimators can estimate over the entire space.

✓ Nice work

5. What classification decision rule is used to classify a new data point when applying density estimation to implement a binary classifier (assume a single input feature)? 1 / 1 point

- ( ) The classifier should estimate the probability of each class given the new data point, and choose the class that has highest prior probability.
- (•) The classifier should estimate the probability of each class given the new data point, and choose the class that is most likely.
- ( ) The classifier should estimate the probability of each class given the new data point, and choose the class that has maximum probability over all values of the feature space.

✓ Nice work

Correct: the classifier will choose the class that has highest posterior class probability.